TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

5-Anchor Subitizing, Regrouping & Base-10 Manipulatives

CARD 01 / 10 • BASE-10

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

5-ANCHOR SUBITIZING & BRIDGING TO 10

Ten-frames visually organize numbers into 5-anchors:
Number = 5 + n and 10 = n + (10 − n)
Seeing 8 as 5 on top and 3 below anchors addition, turning counting into instant visual recognition!

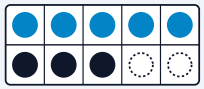
CALIBRATED 14.00 MM SUBITIZING GRID

- Counter Diameter: 14.00 mm (2U) with two-tone face
- Stacking Datum: 1.85mm base plug & stacking socket
- Ten-Frame Grid: 2 × 5 array with 14.00 mm wells
- Base-10 Rods: 10U = 70.00 mm (ten 7mm units)

TWIN 10-FRAME STAGE LID & REGROUPING

Two side-by-side 10-frame stages model 2-digit sums and place-value regrouping across 10 on the lid!

5-ANCHOR SUBITIZING: 5 + 3 = 8 (TEN'S COMP = 2)



INSTANT SUBITIZING

Top Row: 5 Blue (anchor)
Bottom Row: 3 Red (8 total)

2 Empty Slots:

10 − 8 = 2 Ten's Complement
Bridge to 10: 8 + 5 = 13

2. KIT PARTS INVENTORY (34 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
10-Frame Trays	2x	140.0 × 56.0 mm (2×5 Grid)	Matte Charcoal
Two-Color Counters	20x	14.0 mm ⌀ (Blue / Red)	Dual-Sided
Base-10 Rods (10s)	10x	70.0 × 7.00 mm (10U)	Warm Gold
Twin Stage Lid	1x	Side-by-Side 10-Frame	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 32 Manipulatives + Storage Bin + Twin Stage Lid = 34 Parts

DUAL-COLOR HIGH-CONTRAST PALETTE

- Blue Face: First addend or positive value.
- Red Face: Second addend or negative value.
- Warm Gold Rods: 10-unit place-value regrouping rods.
- High-Contrast Wells: Precision visual alignment.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Populate both 10-frames with 10 counters each (all 20 counters accounted for). Lay the 10 gold rods into the center trough. Any empty well instantly spots a piece.
Twin stage lid snaps flush for silent zero-rattle carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

DECIMAL PLACE-VALUE REGROUPING THEOREM

Base-10 positional notation decomposes any integer into powers of 10 with digits 0 through 9.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 01 OF 10 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING • 7.00mm 1U PITCH • 10-FRAME STACKING SLOTS

LESSON GUIDE: MAKE-A-TEN BRIDGE & 5-ANCHOR SUBITIZING

Step-by-Step Hands-On Tutorial: 8 + 5 = 13 via Bridging to 10

Grades K-3 Subitizing & Place Value • Hands-On Classroom & Home STEM Studio Series

CARD 01 / 10 • BASE-10

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/ten-frames/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.1.OA.C.6 (Make Ten)
- CCSS.MATH.CONTENT.2.NBT.B.5 (Place Value)
- TEKS 111.3.b.3.C (Compose/Decompose)

Grades K-3 Subitizing & Place Value

MATHEMATICAL INQUIRY

How many empty spots are left in Frame A before you start moving counters from Frame B?

1 STEP 1: POPULATE 8 ON TEN-FRAME A

Place 8 blue counters in Ten-Frame A.
Fill the top row (5 blue) and bottom row (3 blue).
Instant Subitizing: 5 + 3 = 8 (zero counting needed!).
Notice: exactly 2 slots remain empty to complete the ten.

2 STEP 2: POPULATE 5 ON TEN-FRAME B

Place 5 red counters in Ten-Frame B (fills top row of 5).
We want to calculate 8 + 5 without counting by ones.
Frame A needs 2 counters to bridge to 10.

3-4 STEPS 3 & 4: BRIDGE TO 10 & REGROUP

3. BRIDGE TO 10: Move 2 red counters from B to A. Frame A is now 1 Ten (10)!
4. READ REMAINING ONES: Frame B has 3 red counters left. Total: 10 + 3 = 13!
Keystrokes: Ten-Frame Icon ([■] & [PV]) 8 [ENTER] 2 [+] 5 [+] [■] → 15.

5 CORE PRINCIPLE: ADDITION IS BRIDGING, NOT COUNTING

Counting on fingers by ones is slow and error-prone. The 10-frame proves that addition is decomposing numbers into friendly anchors: 8 + 5 = 8 + (2 + 3) = (8 + 2) + 3 = 10 + 3 = 13!
Once students see the 2 empty spaces in the 8-frame, they instantly know to borrow 2 from the 5!

★ EXTENSION INVESTIGATION: BRIDGE TO 10: 7 + 6

Put 7 in Frame A (3 empty). Put 6 in Frame B. Move 3 to make a ten. How many are left in B? 10 + 3 = 13! (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

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